(1) USB-C J1: data + power / U1: PD sink, requests 9V

CH224A: Rset 6.8k on CFG1 = 9V (datasheet table 5-1). VHV and VBUS pins tied to the connector VBUS, 1uF on VHV.
CC1/CC2 go straight to U1 (Rd is inside U1). DP/DM of U1 are not used (data goes to J4 only).
CFG2/CFG3 = I2C (0x22/0x23, address not selectable) -> own bus per chip. PG is not used (state is read over I2C).
D1 blocks back-feed from VIN to the PC. Shield: 1M || 100nF to GND.
R28 || R29 = 1k bleeder: pulls VBUS1 below 0.8V against the reverse leakage of D1 when only J2 is powered.

J1
TYPE-C-31-M-12
U1
CH224A
C1 1uF 50V
R1 6.8k
R28 2k
R29 2k
R3 1M
C2 100nF 50V
D1 SS54

(2) USB-C J2: power only (charger) / U2: PD sink, requests 9V

Same circuit as (1). D+/D- of J2 are not connected. Shield of J2 goes straight to GND.
D2 blocks back-feed from VIN to the charger port: D1/D2 together pick the higher input.

J2
TYPE-C-31-M-12
U2
CH224A
C3 1uF 50V
R4 6.8k
R30 2k
R31 2k
D2 SS54

VIN bus (4.2 .. 9V)

No TVS: every part on VIN is rated 25V or more
(U3 30V, U4 24V, U6 30V, capacitors 25V).
so a 20V charger fault does not break anything.
Bulk capacitors sit behind D1/D2.

C4 22uF 25V
C5 22uF 25V
TP1 VIN
TP5 GND
PWR_FLAG

U6: 5V LDO (MCU + J4 VBUS)

HT7550-1: 1=GND 2=VIN(tab) 3=VOUT. 10uF in / 10uF out
(datasheet application circuit).
At VIN=5V the output follows VIN minus dropout
(about 4.2 .. 4.9V).

U6
HT7550-1
C6 10uF 25V
C7 10uF 25V
TP4 +5V

U3: 3.3V / 3A buck (TPS62933) + EN from J1 VBUS + JP1 (current measurement)

Vout = 0.8V x (1 + 47k/15k) = 3.307V. RT open = 500kHz. L = 4.7uH (datasheet table 10-2). SS 100nF -> tSS = 100n x 0.8 / 5.5u = 14.5ms.
EN = VBUS1 x 68k/(100k+68k) = 0.405 x VBUS1: turns on at VBUS1 > 3.0V typ (3.16V max). EN = 2.0V @5V, 3.6V @9V, 5.5V @13.6V.
D5 clamps EN to +5V + VF (< 6V abs max) if VBUS1 ever goes higher. MCU (PC3, open drain) can pull EN low. C15 1uF: EN rises in about 37ms.
Charger on J2 alone -> VBUS1 = 0 -> EN low -> 3.3V stays off.

U3
TPS62933DRLR
C8 10uF 25V
C9 10uF 25V
C10 100nF 50V
C11 100nF 50V
C12 100nF 50V
L1 4.7uH FXL0630
R6 47k
R7 15k
C13 22uF 25V
C14 22uF 25V
R8 100k
R9 68k
C15 1uF 50V
D4 1N4148W
R10 1k
LED3 Red
JP1 3V3 current
JP5 Solder jumper (closed)
TP2 +3V3

U4: 12V / 0.5A boost (MT3608B) + Q1/Q2 output switch + JP2 (current measurement)

Vout = 0.6V x (1 + 100k/5.1k) = 12.36V. L = 4.7uH. D4 = SS34 (datasheet application). EN pulled down 100k -> off with a blank MCU.
A boost passes VIN to its output when stopped, so Q1 (AO3401A, Vgs max +-12V) cuts the output. Q2 and Q3 in series pull the gate down
through 100k only when SW12_ON is high AND +3.3V is present: Vgs = -12.36 x 100k/200k = -6.2V (-9.8V if VIN = 20V).
C32 47nF drain-gate slows the edge to about 2 V/ms. Q2 gate pulled down 100k -> off with a blank MCU.
F1 (PTC 0.75A hold) protects against a short on the card. R18 bleeds the +12V line to 0V when off.

U4
MT3608B
C16 22uF 25V
C17 100nF 50V
L2 4.7uH FXL0630
D4 SS34
R11 100k
R12 5.1k
C18 22uF 25V
C19 22uF 25V
R13 100k
Q1
AO3401A
Q2
2N7002
R14 100k
C20 100nF 50V
C32 47nF 50V
R15 100k
R16 100k
JP2
JP2 12V current
F1 nSMD075-13.2V
R18 10k
TP3 +12V

USB 2.0 data: J1 -> U5 (ESD) -> J4

SRV05-4: 1=IO1 2=GND 3=IO2 4=IO3 5=REF 6=IO4.
D+ on IO1, D- on IO2 (one diode pair per line).
IO3, IO4 not used.
REF is left open: its TVS breaks down at 6V.
VBUS1 can be 9V.
J4 = motherboard USB 2.0 header pinout:
1 VBUS, 3 D-, 5 D+, 7 GND, 9 key (pull the pin).
JP3 (solder jumper, closed) feeds +5V to J4 pin 1.

U5
SRV05-4-P-T7
J4
J4 USB header
JP3 Solder jumper (closed)

U7: CH32V003F4P6 — reads CC / VIN / VBUS1, switches 3.3V and 12V

VDD = +5V (4.2 .. 5V). ADC reference = VDD, so firmware corrects with the internal Vref channel.
CC1/CC2: 100k series + 100nF (the 100k keeps the load off the PD line: fault current into the pin < 0.2mA).
VIN / VBUS1 sense = x 33k/(100k+33k) = 0.248: 5V -> 1.24V, 9V -> 2.23V, 20V (TVS clamp) -> 4.96V (below VDD = 5V whenever VIN > 5.5V).
PC3 = EN_3V3 and PC4 = PERST# (drive low or leave Hi-Z, never push-pull high). PD2 = EN_12V, PD3 = SW12_ON (100k pull-downs).
PA2/A0 = +12V sense (x 0.248: 12.36V -> 3.07V).
I2C1 (PC1/PC2) -> U1. Remap I2C1 to PC5/PC6 (AFIO I2C1_RM = 10b) -> U2. No external pull-ups: CH224A has them inside.
J5: 1 = SWIO, 2 = GND (WCH-LinkE). SW1: closed to GND = manual 12V override.

U7
CH32V003F4P6
C21 100nF 50V
C22 1uF 50V
R19 10k
R20 1k
LED5 Red
R21 100k
C23 100nF 50V
R22 100k
C24 100nF 50V
R23 100k
R24 33k
C25 100nF 50V
R25 100k
R26 33k
C26 100nF 50V
R32 100k
R33 33k
C33 100nF 50V
SW1
MSK12C02
J5
J5 SWIO / GND

J3: PCI Express x1 slot — power pins only

Pin names follow the PCIe CEM pinout (side B = component side of the card, side A = solder side).
+12V: A2 A3 B1 B2 B3. +3.3V: A9 A10 B8. 3.3Vaux: B10 via JP4 (solder jumper, closed).
GND: A4 A12 A15 A18 B4 B7 B13 B16 B18. PERST# (A11): 15k to +3.3V, 10uF to GND -> 2.0V about 140ms after 3.3V. D6 empties C31 when 3.3V drops.
PRSNT#, WAKE#, JTAG, SMBus, REFCLK and the lane pins are left open.

J3
PCIe x1 slot
C27 22uF 25V
C28 100nF 50V
C29 10uF 25V
C30 100nF 50V
R27 15k
D6 1N4148W
C31 10uF 25V
JP4 Solder jumper (closed)

Mechanical

M3 mounting holes (not plated,
no net).

H1 M3
H2 M3
H3 M3
H4 M3

Generated by scripts/gen_schematic.py — edit the script, not this file
USB-C x2 -> 3.3V / 12V for USB-data PCIe tuner cards (PX-MLT5PE etc.)

danything	
Sheet: /	
File: pcie-tuner-board.kicad_sch	
Title: PCIe tuner external power board	
Size: A2	Date: 2026-11-11
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